

M.Sc Nestor D. Pereira Neto

ELECTRONIC ENGINEER · 38

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“Graduated in Electronic Engineering (2013) and specializing in Biomedical Engineering (2019). My professional journey includes practical experience in the maintenance of diagnostic imaging hospital equipment, along with a stint in teaching technical and higher education courses since 2011. Recently, I earned a Master’s in Electrical Engineering with a focus on computing and robotics (2022), propelling me into the realm of research and development in robotics since 2019. My expertise spans hardware description, digital signal processing, C/C++ programming, embedded systems, and mobile robotics.”

Education

The University of Manchester

POSTGRADUATE RESEARCH VISITING STUDENT DEPARTMENT PHYSICS AND ASTRONOMY

- Visiting student as part of my PhD studies.

Manchester, GBR

Mar. 2025 - Currently

Federal University of Bahia - UFBA

DOCTORATE DEGREE IN ELECTRICAL ENGINEERING - RESEARCH LINE: MACHINE LEARNING AND DIGITAL SIGNAL PROCESSING

- Development triggered event algorithm based on FPGA for the ATLAS experiment at the Large Hadron Collider (LHC-CERN).

Salvador, BRA

Fev. 2023 - Currently

MASTER’S DEGREE IN ELECTRICAL ENGINEERING - RESEARCH LINE: COMPUTING AND ROBOTICS

- Communication between *Robot Operating System - ROS* and *System-on-a-chip - SoC* with integrated FPGA.

Apr. 2018 - Dec. 2022

Estácio de Sá University

SPECIALIZATION IN BIOMEDICAL ENGINEERING

- Convolutional Neural Network for Detection of QRS Complexes in Electrocardiogram Signals

Salvador, BRA

Oct. 2017 - Feb. 2019

Faculty of Science and Technology - ÁREA1

BACHELOR’S DEGREE IN ELECTRONIC ENGINEERING

- Command prototype to fit X-ray equipment to the standards required by the National Health Surveillance Agency - ANVISA.

Salvador, BRA

Feb. 2008 - Jul. 2013

Professional Experiences

The European Organization for Nuclear Research - CERN

PHD STUDENT

- Development triggered event algorithm based on FPGA for the ATLAS experiment at the Large Hadron Collider.

Geneva, CHE

Jan. 2024 - Currently

Oceaneering

SOFTWARE ENGINEER R&D

- Collaborate with cross-functional teams to ensure seamless integration and functionality of oil and gas systems.
- Software development using modern C++, participate in design reviews, code reviews, and testing to maintain high-quality standards.

Rio de Janeiro, BRA

Dec. 2023 - Feb. 2025

SENAI - CIMATEC

SPECIALIST II - ROBOTIC ENGINEERING

- Development robotics solutions using the ROS2 with expertise in Gazebo Simulation, ROS2 control, SLAM and URDF/Xacro.
- Software and firmware development using C/C++ and python, to ensure integration and functionality of robotic systems.
- Collaborating closely with coworkers guiding trainees and fellows.

Salvador, BRA

Apr. 2022 - Dec. 2023

CONSULTANT II - ROBOTIC ENGINEERING

- Development robotics solutions using the ROS2 with expertise in Gazebo Simulation, ROS2 control, SLAM and URDF/Xacro.
- Software and firmware development using C/C++ and python, to ensure integration and functionality of robotic systems.

Apr. 2019 - Apr. 2022

PROFESSOR PART TIME

- Teach class for the course: Industrial Mechatronics. Subjects: Analog Electronics and Digital Eletronic.

Nov. 2018 - Apr. 2019

ÁREA1|Wyden College

LABORATORY TECHNICIAN

- Prepare and teach practical classes for the courses: Computer Engineering, Electrical Engineering and Automation Engineering.
- Test new tools used in laboratories and provide training, extension courses

Salvador, BRA

Jul. 2013 - Aug. 2017

Alfamed Eletromedicina

ENGINEER - TECHNICAL RESPONSIBLE

- Technical responsible for the company.
- Electronic maintenance in hospital bio-imaging equipment.

Salvador, BRA

Aug. 2013 - Feb. 2015

Languages

Portuguese Native.

English B1 level - CEFR.

Courses

2022 **NR10 - Sistemas Elétricos de Potência - SEP (40)**, IN1000

Salvador, BRA

2022 **Segurança em Eletricidade - NR10(40)**, SENAI

Salvador, BRA

2021 **FPGA INTEL Training(20h)**, Macnica DHW - Official Training Center FPGA INTEL

Florianópolis, BRA

2010 **Study of Device Control via parallel, serial and USB ports (60h)**, Faculdade ÁREA1.

Salvador, BRA

2010 **Applied Analog Electronics (30h)**, Faculdade ÁREA1.

Salvador, BRA

Projects and Research

SENAI CIMATEC

Salvador, BRA

SuBot - CTG BRASIL (CHINA THREE GORGES CORPORATION)

Oct. 2021 - Sep. 2023

- Mobile robotics development project for inspection of high voltage substations.
- Development in C/C++; firmware for peripheral communication; Electrical design.
- Tools: GitHub, ROS2, Gazebo, Mbed, SolidWorks Electrical.

DIGISUB - PETROBRAS

Apr. 2019 - Oct. 2021

- 3D digitizer development project of underwater surfaces in deep water.
- Development in C/C++; firmware for peripheral communication; Electrical design.
- Tools: GitHub, ROS1, Gazebo, PCL, KiCad.

Polytechnic School of the Federal University of Bahia - UFBRA

Salvador, BRA

MASTER'S PROJECT

Dec. 2018 - Dec. 2022

- 1 GigE communication development between FPGA and ROS.
- Development in C/C++ language; Hardware project in verilog for SOC/FPGA; Paper presented at IBERCHIP 2022.
- Tools: Quartus, GitHub, C/C++ Socket, Cyclone V - Intel, Network protocols.

Faculty of Science and Technology - ÁREA1

Salvador, BRA

UNDERGRADUATE RESEARCH SCHOLARSHIP

Jul. 2011 - Jul. 2013

- Development of a microcontrolled command to fit old X-ray equipment to current technical standards established by ANVISA.
- Programming 8 bits microcontrollers; Electronic project; Hardware prototipe; Publication in Cientefico ISSN 1677-1591, jul/dez 2013.
- Tools: 8051, KiCad, Assembly programming language.

Skills and Abilities

Basic Verilog HDL, Cyclone V, Sockets, Cmake, GDB, Nios II, Embedded Linux, SimuLink.

Intermediate Python, 3D CAD, Quartus Prime, RTOS, OOP, Linux, Git/Github, PIC, AutoCAD, ROS/Gazebo.

Advanced PCB/KiCad, Matlab, C/C++, AVR, 8051, ARM Cortex M, GNU-Make, LaTeX.

Presentation

28th IBERCHIP Workshop

Santiago, CHL

IEEE CIRCUITS AND SYSTEMS SOCIETY IN LATIN AMERICA

1-4 Mar. 2022

- Communication between Robot Operating System - ROS and SoC with integred FPGA

Honors & Awards

2022 **PETROBRAS**, Prêmio Inventor 2022 - Award for filing a patent for an underwater 3D scanner

Rio de Janeiro, BRA